

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2023-24

Zoology

Paper: ZOB-403A — Applied Zoology (Ancillary Course)

Time: Three hours

Full Marks: 70

Note: Answer six questions including Question No. 1 which is compulsory. Draw diagrams wherever necessary. The figures in the right-hand margin indicate marks.

Question 1. Answer the following parts:

(a) State whether the following statements are True or False. Justify your answer in 2–3 sentences each: [5 × 2 = 10]

- (i) Karan Swiss is developed at the National Dairy Research Institute of India (NDRI), Karnal by crossing of Brown Swiss and Sindhi breeds of cow.
- (ii) Drone bees are unable to guard the hive against intruders.
- (iii) The GloFish is a genetically modified fish produced by integrating the Green Fluorescent Protein (GFP) gene from firefly into the embryo of Zebrafish (*Danio rerio*).
- (iv) *Streptococcus thermophilus* is used as a starter in curd, yogurt and cheese.
- (v) Lac, the commercial resin of animal origin, is secreted only by the larvae of *Laccifer lacca*.

(b) Define the following: [6 × 1 = 6]

- (i) Mariculture
- (ii) Koinobiont parasitoid
- (iii) Propolis
- (iv) Broiler
- (v) Low-temperature-longer-time pasteurization
- (vi) Stifling.

(c) Differentiate between the following: [2 × 2 = 4]

- (i) Finfish and Shellfish
- (ii) Snake venom and Bee venom.

Question 2. Describe different methods of preservation and processing of prawn. [10]

Question 3. Give a description of social organization in honeybee, focusing on the division of labour in particular. [10]

Question 4. Answer the following parts:

- (a) Write an essay on major disadvantages of using chemicals in controlling of insect pests. [7]
- (b) Write briefly about botanical insecticides. [3]

Question 5. Answer the following parts:

- (a) Illustrate the steps of making transgenic animals with the help of a suitable diagram. [6]
- (b) Discuss snake milking process in brief. [4]

Question 6. Answer the following parts:

- (a) Write an essay on indigenous breeds of bovine. [8]
- (b) What is the composition of milk? [2]

Question 7. Explain the process of sericulture. Add a note on its importance and scope in India. [10]

Question 8. Answer the following parts:

- (a) Write a note about the different difficulties faced by pearl farmers. [4]
- (b) What are the applications for byproducts derived from apiculture? [4]
- (c) Write a note on the nutritive values of chicken. [2]

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